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Where I Becomes Infinite

Zero-Shot Prompting

Module 2.3: Prompt Engineering | Prof. Sashikumaar Ganesan | IISc Bengaluru

Session Overview

What You Will Learn

- What zero-shot prompting is
- Why LLMs can answer without examples
- When zero-shot works and when it fails
- How to write effective zero-shot prompts

Session Details

- **Duration:** 7 minutes
- **Bloom's Level:** B2 — Understand
- **Difficulty:** L1 — Beginner
- **Module:** 2.3 Prompt Engineering
- **Prereq:** Prompt Engineering Principles

Learning Objectives

By the End of This Session You Will Be Able To

- **Define** zero-shot prompting and distinguish it from few-shot
- **Explain** why pre-trained LLMs can generalize to unseen tasks
- **Write** effective zero-shot prompts for classification, QA, and generation tasks
- **Identify** task types where zero-shot is sufficient vs. where it falls short

Zero-Shot Prompting: Definition

Core Definition

Zero-shot prompting is asking an LLM to perform a task by providing only an instruction and input, *with no examples* of how the task should be completed. The model must generalize entirely from its pre-training knowledge.

The “Zero” in Zero-Shot

- **Shot** = demonstration or example provided in the prompt
- **Zero shots** = no examples at all; just instructions and input
- Relies on the model’s ability to understand task intent from language alone
- The model has seen millions of similar tasks during pre-training

Why Zero-Shot Prompting Works

Foundation in Pre-Training

- LLMs are trained on trillions of tokens from books, websites, and code
- They have seen countless examples of summarization, translation, classification, question-answering, and more during training
- Instruction fine-tuning (InstructGPT, RLHF) further aligns models to follow natural language directives without examples
- Modern models (GPT-4, Claude, Gemini) are specifically optimized for zero-shot performance

Key Insight

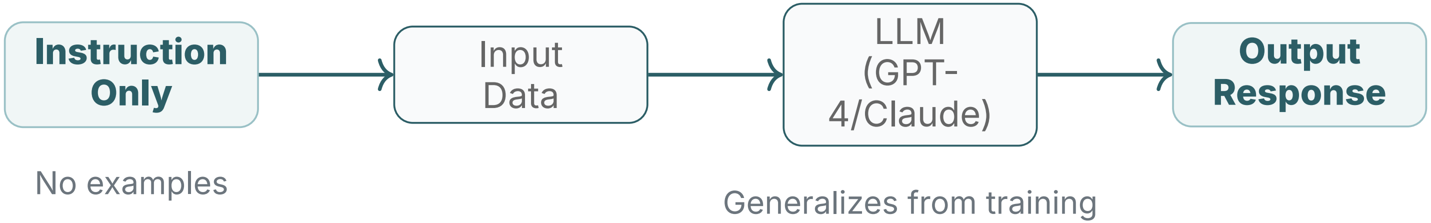
Zero-shot works because the model has already learned the task implicitly. Your instruction simply *activates* knowledge the model already possesses.

Zero-Shot in Action: Text Tasks

Example Prompts

- **Sentiment Classification:** “Classify the sentiment of this review as Positive, Negative, or Neutral. Review: ‘The product arrived late and was damaged.’ ”
- **Translation:** “Translate the following sentence from English to French: ‘The meeting is at 3pm on Monday.’ ”
- **Summarization:** “Summarize the following paragraph in one sentence: [paragraph text]”
- **Named Entity Recognition:** “Extract all person names and company names from: [text]”

Zero-Shot Prompt Flow



What Happens Inside

- The model parses the instruction and maps it to a learned task pattern
- No in-context learning is performed; weights are fixed
- The model's RLHF training guides it to follow instructions accurately

When Zero-Shot Works Well

Strong Performance Scenarios

- **Common NLP tasks:** Summarization, translation, classification, Q&A
- **Well-defined instructions:** Task intent is unambiguous from language alone
- **Large, instruction-tuned models:** GPT-4, Claude 3.5, Gemini 1.5 Pro
- **General-knowledge questions:** Factual recall, explanations, definitions
- **Format conversion:** JSON extraction, markdown formatting, code explanation

When Zero-Shot Falls Short

Performance Gaps

- **Novel or domain-specific formats:** Custom schema the model has not seen
- **Complex multi-step reasoning:** Arithmetic word problems, logical chains
- **Highly specialized domains:** Medical coding, legal clause extraction
- **Ambiguous tasks:** When the desired output style is hard to describe in words

In these cases, providing one or more examples (one-shot or few-shot prompting) significantly improves performance.

Writing Better Zero-Shot Prompts

Practical Guidelines

- Use a clear action verb: classify, extract, translate, summarize, generate
- Specify the output format explicitly: "Return only the label. Do not explain."
- Add a role for domain-specific tasks: "You are a medical coding specialist."
- State constraints upfront: "Answer in under 50 words. Use formal English."
- Separate instruction from input clearly using delimiters or labels

Application: Content Moderation at Scale

Zero-Shot Classification in Production

Platforms such as YouTube, Twitter/X, and Reddit use LLM-based zero-shot classifiers to detect harmful content, spam, and policy violations. A single well-engineered zero-shot prompt can replace complex rule-based systems trained on labeled data.

Example Prompt Pattern

- Instruction: "Classify the following comment as: Safe, Spam, Hate Speech, or Violence."
- Input: The user-submitted text
- Output indicator: "Respond with only one label."
- No labeled training data required for deployment

Application: Enterprise Knowledge Q&A

Internal Search and Helpdesk Automation

Companies deploy zero-shot LLMs on internal documentation to answer employee questions without building labeled datasets. Systems like Glean, Notion AI, and Microsoft Copilot for Microsoft 365 use this pattern daily.

Zero-Shot Q&A Prompt Pattern

- Role: “You are a helpful internal HR assistant.”
- Context: Relevant policy document injected from retrieval
- Instruction: “Answer the employee’s question using only the provided document.”
- Fallback: “If the document does not contain the answer, say: I don’t know.”

Application: Code Explanation and Review

Developer Productivity Tools

GitHub Copilot Chat, Sourcegraph Cody, and JetBrains AI Assistant use zero-shot prompts to explain code, identify bugs, and suggest improvements without needing labeled code-explanation pairs.

Zero-Shot Code Tasks

- "Explain what this function does in plain English."
- "Identify any security vulnerabilities in the following code."
- "Convert this Python function to TypeScript."

Zero-Shot vs. Few-Shot: Quick Comparison

Dimension	Zero-Shot	Few-Shot
Examples in prompt	None	1–10 examples
Prompt length	Short	Longer
Context window use	Minimal	Moderate to high
Best for	Common tasks, clear instructions	Novel formats, complex tasks
Model requirement	Large, RLHF-tuned	Any size
Setup effort	Very low	Requires curated examples

Key Takeaways

Core Concepts to Remember

- Zero-shot prompting provides no examples — only instruction and input
- It works because LLMs learn task patterns implicitly during pre-training
- Instruction-tuned models (GPT-4, Claude, Gemini) are optimized for zero-shot
- It excels at common tasks with clear instructions; struggles with novel formats
- When zero-shot falls short, move to one-shot or few-shot prompting

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